RESPONSIBLE TECHNOLOGY ADOPTION

Literature Review Matrix and Conceptual Framework

Research Title: WEB-TRANSCO: A Web-Based Booking Platform for Jeepney Transport Services

Recommended Technology (Week 7): Cloud Computing | CO2 – Responsible Adoption Evaluation

Part A – Literature Review Matrix

The following matrix synthesizes five credible sources that examine Cloud Computing in relation to cybersecurity, privacy, governance, sustainability, business value, and adoption risks. Each entry identifies key findings and maps them to the responsible adoption of a cloud-based booking platform for jeepney transport groups.

Author / Year	Technology / Issue Studied	Key Finding	Relevance to Organizational Problem	Responsible Adoption Consideration
Mnyakin, M. (2023)	Cloud Computing, AI, and IoT in smart transportation systems	Cloud Computing enables vehicle-to-cloud communication, scalable infrastructure, data analytics, Mobility-as-a-Service platforms, and predictive maintenance. Integration with AI and IoT improves overall transport efficiency, safety, and cost reduction.	Directly supports the need for a scalable, always-available booking platform that can host reservations for private rides, cargo, and shuttle services while allowing future expansion of digital features.	Adopt cloud as the foundational layer for WEB-TRANSCO; plan for future AI/IoT integration only after a stable data platform and governance structures are in place.
Scalable Computing Practice & Experience (2023)	Data security and privacy in cloud-based intelligent transportation applications	Cloud reliance raises security and privacy concerns. Encryption, access controls, threat detection, and privacy-protection measures can safeguard sensitive transport data and user privacy, thereby building stakeholder confidence and encouraging adoption.	Jeepney booking data (customer identities, locations, payment details, trip history) are sensitive. Security failures would erode trust among operators and customers and could halt platform use.	Implement end-to-end encryption, role-based access control, secure payment gateways, and regular security audits from day one. Prioritize compliance with the Philippine Data Privacy Act.
Openvia (2025)	Cloud services in smart mobility: scalability and security for infrastructure management	Cloud platforms provide elasticity for fluctuating demand in mobility systems. Strict cybersecurity protocols (encryption, multi-factor authentication, continuous monitoring) and compliance frameworks (GDPR, ISO 27001) are essential to maintain public trust.	Booking volume for jeepneys can vary sharply by time of day, weather, and local events. Cloud elasticity matches this demand pattern while security measures protect the platform's reputation.	Select cloud providers with proven compliance certifications. Enforce multi-factor authentication for operators and continuous monitoring of the booking system. Educate cooperative staff on basic security practices.
Al-Sharafi et al. / Penn State systematic review (2025)	Cloud Computing adoption for sustainable business operations – antecedents, advantages, and challenges	Cloud adoption is driven by technological, organizational, and environmental factors. Advantages include efficiency and resource optimization that support sustainability; challenges (security, dependency, skills) act as moderating barriers that must be managed.	Jeepney groups seek efficiency gains and lower operating costs while contributing to more organized, less wasteful transport. Understanding adoption barriers helps design a realistic rollout.	Treat challenges (skills gaps, security configuration, vendor dependency) as manageable moderators. Provide training, start with modest cloud resources, and establish clear service-level agreements to reduce risk.
Burlat, M. P. (2026)	JeepniGo – digital booking, real-time tracking, and fare collection for jeepney commuting in the Philippines	A digital platform integrating booking, tracking, and automated fare collection successfully streamlined operations, improved route visibility, and strengthened communication among passengers, drivers, and cooperatives in a Philippine context.	Provides local evidence that digital booking platforms are feasible and beneficial for jeepney groups, aligning closely with WEB-TRANSCO's goals of reducing manual arrangements and expanding service offerings.	Local success stories support adoption feasibility. Still, responsible rollout must include digital literacy support for drivers/cooperatives, inclusive design for low-bandwidth users, and clear data-use policies to maintain trust.

Part B – Conceptual Framework

The framework below illustrates the logical pathway from the identified organizational problem through the recommended technology, the responsible adoption considerations required for ethical and sustainable implementation, and the expected organizational outcomes.



Part C – Framework Explanation

The organizational problem addressed is the inefficiency of manual booking arrangements for jeepney cargo, shuttle, and private-ride services, which produces delays, miscommunication, and lost opportunities for both operators and customers. In Week 7, Cloud Computing was recommended as the most suitable foundational technology because it supplies the scalable, always-available infrastructure required to host a centralized web-based booking platform. Literature supports this choice: Mnyakin (2023) shows that cloud enables Mobility-as-a-Service platforms and vehicle-to-cloud communication; local evidence from JeepniGo (Burlat, 2026) confirms that digital booking systems can streamline Philippine jeepney operations; and systematic reviews highlight efficiency and sustainability gains when adoption barriers are managed. At the same time, sources emphasize that responsible adoption requires robust cybersecurity (encryption, access controls, continuous monitoring), strict privacy compliance with the Philippine Data Privacy Act, inclusive design that does not exclude low-literacy or low-bandwidth users, clear governance policies and service-level agreements, and attention to the environmental footprint of cloud resources. When these considerations are addressed, the expected outcomes include increased booking volume, reduced operational friction, expanded service offerings, higher customer trust, and stronger long-term competitiveness for jeepney transport groups.

Sources

1. *Mnyakin, M. (2023). Applications of AI, IoT, and Cloud Computing in smart transportation: A review.* Artificial Intelligence in Society. https://www.researchberg.com/index.php/ai/article/view/108
2. *Protecting data and privacy: Cloud-based solutions for intelligent transportation applications. (2023).* Scalable Computing: Practice and Experience, 24(3). https://scpe.org/index.php/scpe/article/view/2381
3. Openvia. (2025). The role of cloud services in smart mobility: Scalability and security for infrastructure management. https://www.openvia.io/the-role-of-cloud-services-in-smart-mobility-scalability-and-security-for-infrastructure-management/
4. *Al-Sharafi, M. A., et al. (2025). Unlocking the potential of cloud computing for sustainable business operations: A systematic review of antecedents, advantages and challenges.* Penn State / Web of Science & Scopus systematic review. https://pure.psu.edu/en/publications/unlocking-the-potential-of-cloud-computing-for-sustainable-busine/
5. *Burlat, M. P. (2026). JeepniGo: A smart mobility platform for accessible, inclusive, and sustainable jeepney commuting in the Philippines.* SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6620819

Note: *This Week 8 activity continues directly from the Week 7 Technology Evaluation Matrix, which recommended Cloud Computing as the most appropriate technology for WEB-TRANSCO. The literature review and conceptual framework focus on ensuring that adoption is not only technically suitable but also responsible, secure, ethical, sustainable, and properly governed.*